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I am a second-year PhD student at the Indian Institute of Science (IISc), Bangalore. Till date, I have done research on *machine learning applications in speech processing* and *robust learning under label noise*. In my research, I seek to understand and solve theoretical problems in *deep learning (DL)* and, more broadly, in *weakly supervised learning* which naturally arise from real-world concerns such as safety and security. For instance, I want to understand (but not limited to) the following: What relations exist between robustness under weak supervision, overparameterisation, and the geometry of loss landscape? What are the tradeoffs between robustness under weak supervision and overparameterisation under fairness/privacy/information constraints that are ubiquitous in, say, federated learning? Can causal inference methods help obtain robustness under weak supervision? My work on real-world hospital data, and a focused investigation of problems like *learning under label noise*, have provided me with hands-on experience and a strong foundation in classical machine learning theory to tackle issues of weakly supervised learning. Moreover, for my future work, these past experiences will help me bring a different perspective to the theory of deep learning: Traditionally, noise in any system is regarded as unwanted, a source of inefficiency, or something to be 'tolerated'. Characterising and understanding different types of 'noise' (noisy supervision signals) as seen in weakly supervised learning paradigms and their effect on deep networks, analogous to classical signal processing approaches, can provide a more principled understanding of complex systems like deep networks.

Foray into research. My first research exposure was at Prof. Prasanta Kumar Ghosh's SPIRE lab, IISc. I worked on automated disease onset and severity prediction for Amyotrophic Lateral Sclerosis (ALS) disease. Speech impairment and its severity are one of the commonly used diagnostic aids since no biomarkers are currently known for ALS [1]. So, to better understand speech's role in ALS, I collected and analysed patients' articulatory movements and audio recordings for various speech tasks. I also co-designed these speech tasks to account for varying levels of speech degradation at different stages of ALS. Due to the underlying medical condition, the data collection process is often arduous for the patients, and their articulatory data can be too noisy to analyse. I analysed the efficacy of various speech tasks for automated prediction using i.) articulatory *and* audio data and ii.) only audio data. We showed that, even without articulatory data, audio data alone can help such prediction tasks, ultimately reducing data collection time and strain on patients. These works were published in *ICASSP 2018 [2]* and *Interspeech 2019 [3]*. This was an instructive experience as it introduced me to two of the many fascinating problems of weakly supervised learning: i.) Doctors gave inconsistent disease-severity labels (i.e. labelling errors), and difficulty in convincing subjects for data collection led to an imbalance in the number of healthy subjects and patients. This is a case of *learning under label noise with class imbalance*; ii.) The hospital we worked with typically receives patients from the southern and eastern states of India. For deployability, any model trained on such data should perform well in spite of the unseen country-wide distributional variations in the training data. This requires *out-of-distribution generalisation*.

Research in Robust Deep Learning. Thereafter, I joined Prof. P S Sastry's lab at IISc as a Master's student to pursue research on some of these topics. Supervised learning methods are formulated under the classical assumption that the training and test data come from the same distribution. However, deep networks are typically trained using large-scale labelled training data, which is often generated through crowd-sourcing or automated labelling, thereby inevitably containing labelling errors. Thus, the aforementioned classical assumption is violated since we want the trained models to perform well only on the test set (which has clean labels) and not on this training set (which has noisy labels). Since deep networks can memorise training data even when trained with noisily labelled data [4], several methods have been proposed to curb this memorisation. These methods often need complex algorithmic modifications, thus affecting the deployability of the DL models. To bypass such complications, I sought to obtain *inherent* robustness to memorisation in the first part of my master's thesis. In particular, because the local minima that networks can reach depend on the loss landscape geometry (corresponding to the empirical risk being minimised), I investigated the effects of changing only loss function during training to achieve this *inherent* robustness. I empirically showed that while standard losses like cross-entropy (CCE) and mean squared error (MSE) result in *memorisation*, special losses, namely, *symmetric losses* (e.g. Robust Log Loss [5]), do not. We formally defined *robustness to memorisation*, and showed that symmetric losses are provably robust to such memorisation. This work was published in *PAKDD 2021 [6]*. This

work has been invaluable in introducing me to the possibility of achieving *inherent* robustness and engineering loss landscape geometry via loss function design for 'robust' learning.

Subsequently, these findings led me to think about whether standard losses (like CCE and MSE) themselves have local minima that can provide such 'robustness'. Since loss function design lacks principled theory, designing algorithms to reach such minima (if they exist) with just standard losses will be of immense practical value. Towards this, in the second part of my master's thesis, my advisor and I proposed an adaptive sample reweighting scheme. Stemming from our observation that the sample loss values change differently for each class and throughout the network's training, we sought to use uncertainty estimates of these values for robustness to memorisation. We conjectured and demonstrated empirically that using only class-wise statistics of CCE loss values helps effectively identify which sample's labels are corrupted, thus indicating that our algorithm is indeed able to reach near such 'robust' minima. My contributions included co-designing the proposed algorithm as a curriculum learning-like strategy and implementing it along with the baselines on my own. Our algorithm is also ~2.5 times faster than some of the best baselines. This work has been accepted in *WACV 2023 [7]*. To understand our algorithm better, I am currently using distribution-independent uncertainty quantification methods, such as conformal prediction, to ask: How to assess the quality of our batch statistics-based estimate with only finite-sample sizes and without any assumption about the class-wise loss value distributions? I am currently also attempting this theoretical analysis through the lens of curriculum learning theory and non-parametric hypothesis testing for separating subsets belonging to two different populations (given finite sample sizes).

Why UW-Madison (UWM)? Research at UWM spans a much broader set of Machine Learning areas, and I believe this will help me pursue my research goals for weakly supervised learning and DL theory at a broader and deeper level. Regarding this, I would like to list some relevant, ongoing research at UWM: i.) *Prof. Xiaojin Zhu's* research on machine teaching and adversarial robustness excites me as it aligns with my research interests. For instance, it would be interesting to use ideas from machine teaching to identify 'good' quality and reliable data – in a manner similar to [8] – for robust learning under different weak supervision paradigms. ii.) *Prof. Sharon Yixuan Li's* research on out-of-distribution generalisation (e.g. [9, 10]) directly aligns with my research interests. iii.) *Prof. Yingyu Liang's* research on generalisation in neural networks, adversarial robustness and out-of-distribution generalisation directly aligns with my interests. iv.) *Prof. Yong Jae Lee* and *Prof. Frederic Sala's* research on weakly supervised learning (e.g. [11, 12, 13]) directly aligns with my interests.

Why PhD? My long-term goal is to join academia so that I can: i.) work towards a theory of deep learning with an initial emphasis on weakly supervised learning, given its practical relevance in an increasingly automated world; ii.) continue to pursue the art of teaching. While pursuing my Master's here at IISc, I developed a strong interest in teaching. I have been a teaching assistant for graduate-level probability and machine learning courses (classes of > 70 students) on campus almost every semester since 2020. I am also a core team member of on-campus, student-run volunteer group, NoteBook Drive (NBD), through which I, along with my team, organise a weekly science mentorship programme, wherein we inculcate scientific temperament in local government school kids through interactive and inquiry-driven sessions on science and maths. Recently, I also helped organise my department's inaugural Summer School in an effort to share the department's research with university students across the country. These experiences have taught me not only the importance of science communication but also improved my ability to do so with audiences from diverse backgrounds. These experiences have also been instrumental in refining my own study practices and improving my grades over the last two years. Doing a PhD helps me pursue research in my interest areas and create and share an environment full of empathetic learning and mentorship experiences with my future collaborators and students.

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